INCH-POUND

MIL-S-24796(SH)
22 June 1995

MILITARY SPECIFICATION

SENSORS, ABSOLUTE PRESSURE, FIBER OPTIC (NAVAL SHIPBOARD USE)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for fiber optic absolute pressure sensors for Naval ships, but does not include the readout display (see 6.1).

1.2 Classification (see 6.2).

1.2.1 Designations. The sensor classification shall consist of a series of designations which shall be assigned and listed in the format below.

24796	- X	- X	- XXX	- XX	- X	- X
Specification	Type	Application	Range	Input Power	Opto-Electronics Module	Output
	1.2.1.1	1.2.1.2	1.2.1.3	1.2.1.4	1.2.1.5	1.2.1.6

1.2.1.1 Type. The sensor type shall be designated by the single letter symbol as follows:

R - Pressure (absolute)

1.2.1.2 Application. The media to be measured shall be indicated as follows:

A - Freshwater, condensate, steam, nitrogen, and other inert gases; and liquid hydrocarbons (fuel, lube oil)
S - Seawater
G - Flue gas and ammonia
O - Oxygen

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 03R42, 2431 Jefferson Davis Hwy., Arlington, VA 22242-5160, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1.3 Range. The pressure range of the sensor head shall be designated as follows:

32B - 14.7 to 30 psia
33B - 14.7 to 60 psia
34B - 14.7 to 100 psia
35B - 14.7 to 200 psia
36B - 14.7 to 300 psia
37B - 14.7 to 400 psia
38B - 14.7 to 600 psia
39B - 14.7 to 1000 psia
40B - 14.7 to 1500 psia

1.2.1.4 Electrical power supply. The opto-electronics module interface wiring shall be determined by the power supply as follows:

AC - Four wire system used with a 115 Volt (nominal) alternating current (ac) supply
DC - Two wire system used with a 28 Volt (nominal) direct current (dc) supply

1.2.1.5 Opto-electronics module. The opto-electronics module shall be designated as follows:

A - Bulkhead mounted
B - Console mounted

1.2.1.6 Output. The electrical signal output of the sensor shall be designated by the following symbols (see 3.4.7).

A - Milliampere (mA)
V - Volt (V)
O - Optical (and current)
D - Digital (and current)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

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SPECIFICATIONS

MILITARY

MIL-S-901	- Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MS3452	- Connector, Receptacle, Electric, Box Mounting, Rear Release, Crimp Contact, AN Type With.
MS3456	- Connector, Plug, Electrical, Rear Release, Crimp Contact, AN Type.
MIL-C-5015	- Connector, Electrical, Circular Threaded, AN Type, General Specification for.
MS16142	- Boss, Gasket Seal Straight Thread Tube Fitting, Standard Dimensions for.
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.
MIL-S-19500	- Semiconductor Devices, General Specification for.
MIL-S-19622	- Stuffing Tubes, Nylon; and Packing Assemblies; General Specification for.
MIL-S-19622/1	- Stuffing Tubes, Straight, Nylon.
MIL-S-19622/17-	Stuffing Tube, Nylon, Size 2: Packing Assemblies for.
MIL-E-24142	- Enclosures for Electrical Fittings and Fixtures, General Specification for.
MIL-E-24142/3	- Enclosure for Electrical Fittings and Fixtures, Submersible, Size 6 by 9 (15-Foot).
MIL-M-24794	- Material, Index Matching, Fiber Optics.
MIL-F-49291	- Fiber, Optical, (Metric), General Specification for.
MIL-P-83461	- Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Improved Performance at 275°F (135°C)
MIL-P-83461/1	- Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Improved Performance at 275°F (135°C) Sizes and Tolerances.
MIL-C-83522	- Connectors, Fiber Optic, Single Terminus, General Specification for.
MIL-C-83522/16-	Connector, Fiber Optic, Single Terminus, Plug, Adapter Style, 2.5 Millimeter Bayonet Coupling, Epoxy.
MIL-C-83522/17-	Connector, Fiber Optic, Single Terminus, Adapter, 2.5 Millimeter Bayonet Coupling, Bulkhead Panel Mount.
MIL-C-83522/18-	Connector Fiber Optic, Single Terminus, Adapter, 2.5 Millimeter Bayonet Coupling PC Mount.

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MIL-C-85045 - Cable, Fiber Optic, (Metric), General Specification for.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-461 - Requirements for the Control of Electromagnetic Interference Emission and Susceptibility.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-701 - Lists of Standard Semiconductor Devices.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-1861 - Electrical and Electronic Assemblies, Boards, Cards, and Associated Hardware, Selection and Use of.
MIL-STD-2042 - Fiber Optic Topology Installation, Standard Methods for Naval Ships.

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY OF TESTING MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to American Society of Testing Materials, 1916 Race Street, Philadelphia, PA 19103-1187)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-422 - Electrical Characteristics of Balanced Voltage Digital Interface Circuit.

(Application for copies should be addressed to Electronic Industries Association, 2001 Eye Street, NW, Washington, DC 20006.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein , the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Reliability. The reliability of the opto-electronics module, expressed in mean time between failures (MTBF), shall be no less than 40,000 hours. A reliability prediction for the opto-electronics module, in the form of an MTBF calculation, shall be performed. The computational techniques specified in MIL-HDBK-217 should be used for guidance.

3.2 First Article. When specified (see 6.2) and prior to production, a sample shall be subjected to first article inspection (see 6.5), in accordance with 4.4.

3.3 Materials.

3.3.1 Metals. Unless otherwise specified herein, all metals used in the construction of the sensor shall be corrosion-resistant and shall be in accordance with MIL-STD-454, requirement 15. Dissimilar metals shall not be used in intimate contact with each other unless suitably finished to prevent electrolytic corrosion and shall be in accordance with MIL-STD-454, requirement 16. The metals used in construction of the wetted parts shall be in accordance with table I.

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Table I. Materials of wetted parts.

Wetted Parts	Service Fluid			
	Fresh Water, Condensate, Steam, Nitrogen, and Other Inert Gases; Liquid Hydrocarbons (e.g., lube oil, fuel)	Seawater	Flue Gas and Ammonia	Oxygen
CRES 304L 316L 321	X	---	X	---
CRES 15-5 PH 17-4 PH 17-7 PH	X	---	---	---
Monel and K-Monel	X	X	---	X
Inconel 600 625 718 750	X	X	---	---
Hastelloy C276	X	X	X	---
Titanium CP 6Al-4V	X	X	---	---
CuNi 70/30	X	X	---	X
Ni Span	X	---	---	---
Tantalum	X	X	---	---

3.3.2 Flammable materials. Materials used in the construction of the sensor shall in the end configuration be noncombustible or fire retardant in the most hazardous conditions of atmosphere, pressure, and temperature to be expected in the application. Fire retardant additives may be used provided they do not adversely affect the specified performance requirements of the basic materials. Fire retardance shall not be achieved by use of nonpermanent additives to the basic material.

3.3.3 Fungus resistant materials. Materials used in the construction of the sensor shall be in accordance with MIL-STD-454, requirement 4. Materials that are not in accordance with requirement 4 of MIL-STD-454 shall be in accordance with grade I classification of MIL-STD-810, method 508.

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3.3.4 Solvents, adhesives, and cleaning agents. If any chemicals or cements are used in bonding of internal sensor components, no degradation shall result during in-service use. MIL-STD-454, requirement 23 may be used as guidance for epoxy and adhesives used in construction of the sensor.

3.3.5 Refractive index matching gels, fluids, or compounds. Refractive index matching gels, fluids, or compounds shall be in accordance with MIL-M-24794.

3.4 Design and Construction. The equipment shall convert a fluid pressure to a continuous linear proportional analog electrical dc signal. The equipment, herein referred to as the sensor, shall consist of a sensor head, opto-electronics module, and connectorized fiber optic cable. Each component of the sensor shall be interchangeable with the appropriate replacement component with respect to form, fit, and function; with the required accuracy specified in 3.5.1 being maintained. The sensor shall be self-referenced to eliminate errors in indication due to cable bending and connector variations.

3.4.1 Sensor head. The fiber optic sensor head shall detect the fluid pressure. The sensor head shall be passive, that is, pressure measurement shall be accomplished through a change in optical properties. Neither electrical nor electronic components shall be used in the construction of the sensor head. The sensor head shall be connectorized, that is, there shall be no cable pigtails.

3.4.1.1 Sensor head configuration. The sensor head shall be constructed in accordance with figure 1. Access to the interior of the sensor head shall be possible in order to replace either the connectorized cable and sensor head components, or both. Replacement of cable or sensor head components shall be accomplished with the sensor body remaining fixed in place at mounting plate and pressure connection points.

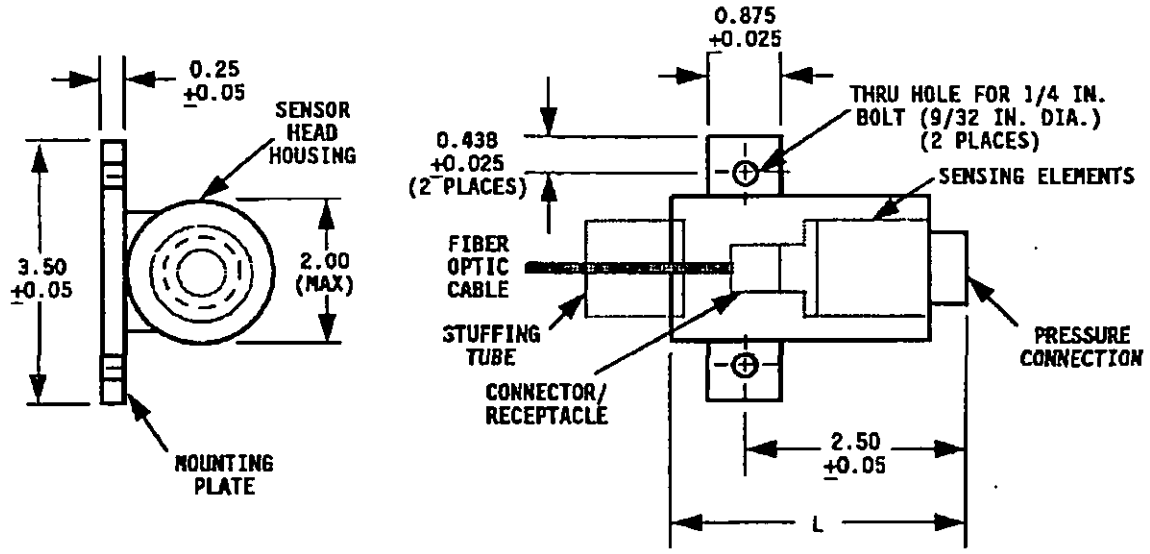
3.4.2 Opto-electronics module. The opto-electronics module shall contain an optical source and detector, and a signal conditioner to convert the sensor output to a continuous linear proportional analog electrical dc signal. Access to the interior of the opto-electronics module shall be possible in order to replace the connectorized cable. The module shall be bulkhead mounted or console mounted as specified in 3.4.2.1 or 3.4.2.2 respectively.

3.4.2.1 Bulkhead mounted (Type A). The opto-electronics shall be housed in a junction box in accordance with MIL-E-24142 and MIL-E-24142/3 (see 6.2).

3.4.2.2 Console mounted (Type B). The opto-electronics shall be packaged in a console mounted circuit card as specified (see 6.2).

3.4.3 Fiber optic cable. For integrity, fiber optic cables shall have an outer diameter of a 4 fiber cable in accordance with MIL-C-85045. In the cable there shall be no less than two times the number of fibers needed for operation of the sensor. The fiber optic cable shall be used to connect the sensor head to the opto-electronics module. The cable shall be supplied with a stuffing tube (size 2, military part number M19622/1-002 in accordance with

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NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. SENSOR HEAD HOUSING (BODY) CROSS-SECTION IS SHOWN AS A CIRCLE. ANY ALTERNATE CROSS-SECTION NOT EXCEEDING 2.00 INCHES IN WIDTH AND 2.00 INCHES IN HEIGHT, MEASURED FROM THE MOUNTING PLATE, IS ACCEPTABLE.
3. 3.00 INCHES $\leq$ L $\leq$ 4.00 INCHES.
4. SHADED ITEMS SHOWN FOR REFERENCE ONLY; NOT INTENDED TO SHOW DESIGN DETAIL.

FIGURE 1. Sensor head configuration.

MIL-S-19622/1) and packing assembly (military part number M19622/17-0001 in accordance with MIL-S-19622/17) in accordance with MIL-S-19622, and an o-ring (military part number M83461/1-214 in accordance with MIL-P-83461/1 and MIL-P-83461) installed on each end of the cable to accomplish watertight penetration into the sensor head and opto-electronics module. The required length of cable shall be as specified (see 6.2). Exposed single fiber optical fiber cable component shall not be used over distances greater than one meter.

3.4.4 Optical fiber. All optical fiber used in the construction of the sensor shall be in accordance with MIL-F-49291.

3.4.5 Fiber optic connectors, receptacles, and bulkhead adapters. All fiber optic connectors, receptacles, and bulkhead adapters shall be in accordance with MIL-C-83522 and MIL-C-83522/16, 17, and 18, respectively. Connectors shall be assembled at both ends of the fiber optic cable between the sensor head and the opto-electronics module. MIL-STD-2042 method 5B1 may be used as guidance. The connectors and receptacles shall be mounted inside the sensor head and opto-electronics unit.

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3.4.6 Electrical input power requirements. Nominal steady-state power supply requirements for ac shall be 115 ± 8 volts, 60 ± 2 Hertz (Hz), single phase. Nominal steady-state power supply requirements for dc shall be 28 ± 4.5 volts. The sensor shall meet all performance requirements specified herein while operating with power supply variations as specified in 4.7.7 and 4.7.8. Full-time surge protection shall be provided for the above power supply limits.

3.4.7 Output. The electrical signal output of the sensor shall be dc directly proportional to the pressure input. The output shall be a true current source of 4 to 20 milliamperes (mA) dc or a true voltage source of 0 to 5 volts (V) dc, as specified (see 6.2). If an optical or digital output is required, it shall be a supplemental output to a true current output (see 3.4.7.1) and shall be as specified in 3.4.7.3 and 3.4.7.4 respectively.

3.4.7.1 Current output. When a 4 to 20 mA current output is specified, the requirements specified herein shall be met regardless of external load resistance variations over a range of 0 to 250 ohms. The 4 mA output shall correspond to the lower pressure range value and the 20 mA output shall correspond to the upper pressure range value for the ranges specified in 1.2.1.3.

3.4.7.2 Voltage output. When a 0 to 5 volt output is specified, the requirements specified herein shall be met for external load resistances exceeding 100 thousand ohms. The 0 volt output shall correspond to the lower pressure range value and the 5 volt output shall correspond to the upper pressure range value for the ranges specified in 1.2.1.3.

3.4.7.3 Optical output. If an optical output is required, the optical output requirements shall be as specified (see 6.2).

3.4.7.4 Digital output. When an electrical digital output is required, the digital output requirements shall be as specified. The electrical characteristics shall be in accordance with EIA standards RS-422 for balanced voltage digital interface circuitry, or as specified (see 6.2). The data format shall be as specified (see 6.2).

3.4.8 Electrical connectors. An electrical interface connector receptacle and mating plug, in accordance with MIL-C-5015 shall be provided with each opto-electronics module of the sensor. The electrical receptacle may be designed to include built-in radio suppression elements. Electrical connectors for digital output shall be as specified (see 6.2).

3.4.8.1 AC power input - current output. The receptacle shall be an MS3452W14S-5PX in accordance with MS3452. Receptacle pins "A" and "B" shall be 115 Vac power input, pin "C" shall be case ground, pin "D" shall be positive 4-20 mA dc signal output, and pin "E" shall be negative 4-20 mA dc signal output. The mating plug shall be an MS3456W14S-5SX in accordance with MS3456.

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3.4.8.2 AC power input - voltage output. The receptacle shall be an MS3452W14S-5PX in accordance with MS3452. Receptacle pins "A" and "B" shall be 115 Vac power input, pin "C" shall be case ground, pin "D" shall be positive 0-5 Vdc signal output, and pin "E" shall be negative 0-5 Vdc signal output. The mating plug shall be an MS3456W14S-5SX in accordance with MS3456.

3.4.8.3 DC power input - current output. The receptacle shall be an MS3452W14S-5P in accordance with MS3452. Receptacle pin "A" shall be positive 28 Vdc power input, pin "B" shall be negative 28 Vdc power input, and pin "C" shall be case ground. Receptacle pins "A" and "B" shall also serve as the 4-20 mA signal output. The mating plug shall be an MS3456W14S-5S in accordance with MS3456.

3.4.8.4 DC power input - voltage output. The receptacle shall be an MS3452W14S-5P in accordance with MS3452. Receptacle pin "A" shall be positive 28 Vdc power input, pin "B" shall be negative 28 Vdc power input, pin "C" shall be case ground, pin "D" shall be positive 0-5 Vdc signal output, and pin "E" shall be negative 0-5 Vdc signal output. The mating plug shall be an MS3456W14S-5S in accordance with MS3456.

3.4.9 Pressure connections. Pressure sensing connections for all services shall be a 7/16-20 UNF-2B tube connection in accordance with MS16142.

3.4.10 Adjustments. Tamper-proof adjustments for zero and span shall be provided on the opto-electronics module for calibration purposes. The number of adjustments shall be kept to a minimum consistent with the operation and maintenance requirements. Electrical disconnection shall not be required to accomplish these adjustments.

3.4.11 Printed circuit boards. Printed circuit boards shall be in accordance with MIL-STD-1861.

3.4.12 Fuses. The opto-electronics module shall not be fused.

3.4.13 Semiconductors. Silicon rectifiers, diodes, photodiodes, transistors, and other semiconductor devices shall be as specified in MIL-S-19500 and shall be in accordance with MIL-STD-701.

3.4.14 Weight. The weight of the sensor head shall be not greater than 18 ounces. The weight of the opto-electronics module shall be not greater than 10 pounds.

3.4.15 Sensor mounting. The sensor head shall be mounted using a mounting plate as shown on figure 1. If required in a specific application and with prior approval of NAVSEA, the sensor head may be mounted by its pressure piping connection. If the sensor head is mounted by its pressure connection, the mounting plate may be removed (see 6.2).

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3.4.16 Calibration media. Oil shall not be used as the calibration media. The legend "USE NO OIL FOR CALIBRATION" shall be prominently marked on the sensor housing.

3.4.17 Welding. For application 0 (see 1.2.1.2), all internal pressure containing parts shall be joined by welding according to MIL-STD-278.

3.4.18 Lubrication. The sensor shall not require lubrication. The legend "DO NOT LUBRICATE" shall be prominently marked on the sensor housing.

3.4.19 Damping. The use of oil for damping is prohibited.

3.5 Performance requirements.

3.5.1 Accuracy. The accuracy of the sensor shall be within plus or minus 1 percent of span including thermal effect over the temperature range of 0°C to 65°C (see 4.7.1). Span is the algebraic difference between the limits of the range.

3.5.2 Repeatability. The repeatability of the sensor output shall be within 0.5 percent of span (see 4.7.1).

3.5.3 Sensitivity factor. The values of the ratio of electrical output percentage change to pressure input percentage change shall not be less than 0.75 or more than 1.25 (see 4.7.3).

3.5.4 Response. Sensor output shall conform to the following criteria (see 4.7.4).

(a) Sensor output shall be within plus or minus 2 percent of span of maximum ramp pressure within 0.01 second of the time that pressure is attained.
(b) Sensor output shall exhibit no overshoot of maximum ramp pressure greater than 2 percent of span.
(c) Sensor output shall indicate the actual pressure to within plus or minus 1 percent of span in not greater than 0.2 second after attainment of maximum ramp pressure, and shall remain within this error band for the duration of applied steady-state pressure.

3.5.5 Warm-up time. Sensor output shall attain a value within plus or minus 1 percent of the steady-state output with no overshoot greater than 1 percent of span. Output shall reach this band in not greater than 1 minute after the sensor is energized and shall remain in this band (see 4.7.5).

3.5.6 Ripple. Peak-to-peak output ripple shall be not greater than 0.5 percent of span (see 4.7.6).

3.5.7 Steady-state supply voltage and frequency (for ac-powered sensors) or steady-state supply voltage (for dc powered sensors). For ac-powered sensors, the difference between outputs at any voltage and frequency condition and the

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normal (115 Vac, 60 Hz) at the same input and test temperature shall be not greater than 1 percent of span (see 4.7.7). For dc-powered sensors, the difference between outputs at any voltage condition and the normal (28 Vdc) at the same input and test temperature shall not be greater than 1 percent of span (see 4.7.7).

3.5.8 Transient supply voltage and frequency (ac) or supply voltage (dc) (see 4.7.8).

3.5.8.1 Voltage. During the voltage transient test, the sensor output shall remain within plus or minus 2 percent of span of the pretransient output.

3.5.8.2 Frequency. During the frequency transient test, the sensor output shall remain within plus or minus 0.5 percent of span of the pretransient output.

3.5.9 Insulation resistance. The insulation resistance of the sensor shall not be less than 10 megohms (see 4.7.9).

3.5.10 Power interruption. During periods when the sensor is energized during the power interruption test (see 4.7.10), sensor performance shall conform to 3.5.1.

3.5.11 Short circuit. The opto-electronics module shall be subjected to a short circuit test (see 4.7.11). The sensor shall exhibit no damage and shall conform to 3.5.1.

3.5.12 Line voltage reversal (dc). The sensor shall conform to 3.5.1 after the line voltage reversal test (see 4.7.12). There shall be no evidence of visual damage to the opto-electronics module as a result of the line voltage reversal test.

3.5.13 Temperature. During the temperature test, sensor performance shall conform to 3.5.1 (see 4.7.13).

3.5.14 Humidity. Deviation of the second cycle reference measurement from the post-conditioning reference measurement shall be not greater than 1 percent of span. After temperature cycling, performance shall conform to 3.5.1 and shall show no evidence of damage (see 4.7.14).

3.5.15 Enclosure. There shall be no evidence of water leakage into the opto-electronics module (see 4.7.15).

3.5.16 Salt spray. No appreciable corrosion or other damage; either optical, mechanical, or electrical; shall be evident after the salt spray test. After completion of the test, performance shall conform to 3.5.1 (see 4.7.16).

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3.5.17 Overpressure. Calibration conducted after the overpressure test shall have no values greater than 1 percent of span deviation from the preoverpressure test calibration. After calibration, sensor performance shall conform to 3.5.1 (see 4.7.17).

3.5.18 Pressure cycling. Calibration conducted after completion of the pressure cycling test shall have no values greater than 1 percent of span deviation from pretest reference measurement (see 4.7.18).

3.5.19 Vibration. The sensor output shall be monitored throughout the test and shall be not greater than plus or minus 2 percent of span deviation from steady-state. There shall be no visible evidence of damage to the sensor as a result of the vibration test (see 4.7.19).

3.5.20 Shock. The sensor shall operate during and after the shock test. After the shock test, but before any adjustment is made, the sensor output shall show no deviation greater than 3 percent of span. After the post test calibration, sensor performance shall conform to 3.5.1. There shall be no visual evidence of damage to the sensor as a result of the shock test (see 4.7.20).

3.5.21 Burst pressure. The sensor shall withstand the burst pressure specified in 4.7.21 without showing any evidence of leakage.

3.5.22 Electromagnetic interference (EMI) emission and susceptibility. The sensors shall be in accordance with the following requirements of MIL-STD-461: CE101, CE102, CS101, CS114, CS116, RE101, RE102, RS101, RS103 and RS105 (see 4.7.22).

3.6 Cleaning. Prior to packaging for delivery, the contractor shall ensure that pressure sensors shall be free of all loose scale, rust, grit, filings, mercury, oil, grease, and other foreign substances or organic materials. The surfaces to be cleaned shall be the pressure connection and all parts contacted by the measurement fluid, except for oxygen where "clean" shall include all surfaces of the sensor. In addition, the following shall apply.

(a) Sensors for application O, oxygen service (see 1.2.1.2), shall be clean gas calibrated, cleaned, and the pressure connections capped.
(b) Sensors for all other applications shall be fresh water or clean gas calibrated, cleaned, and the pressure connections capped.

3.7 Surface finishes. Surfaces of castings, forgings, molded parts, stampings, and machined and welded parts shall be free of defects such as cracks, pores, undercuts, voids, and gaps. External surfaces shall be smooth and edges shall be either rounded or beveled. There shall be no burn-through, warpage, or dimensional change due to heat from welding. There shall also be no damage to adjacent parts resulting from welding.

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3.8 Identification marking. The sensor shall be permanently and legibly marked in accordance with MIL-STD-130. At a minimum, the following information shall be provided.

3.8.1 Sensor head.

(a) Manufacturer's name.
(b) Manufacturer's serial number.
(c) Manufacturer's part number and drawing number.
(d) Government designation (see 1.2.1).
(e) National Stock Number (NSN).
(f) "USE NO OIL FOR CALIBRATION".
(g) "DO NOT LUBRICATE".

3.8.2 Opto-electronics module.

(a) Manufacturer's name.
(b) Manufacturer's serial number.
(c) Manufacturer's part number.
(d) Electrical and optical (if any) output (see 3.4.7).
(e) NSN.
(f) Technical manual number.
(g) Contract number.

3.9 Labeling. A visible label shall be affixed to the outside of the opto-electronics module cover and shall contain the following.

NOTICE
UNTERMINATED OPTICAL CONNECTIONS MAY
EMIT LASER RADIATION
DO NOT VIEW BEAM WITH OPTICAL INSTRUMENTS
AND AVOID DIRECT EXPOSURE TO THE BEAM.

A visible label shall be affixed to the sensor head and the inside of the opto-electronics module and shall contain the following.

WARNING
INVISIBLE LASER RADIATION
AVOID EXPOSURE TO THE BEAM

The labels shall be yellow lettering on a black background.

3.10 Workmanship. Workmanship of the sensors shall be in accordance with MIL-STD-454, requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein.

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Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Standard operating conditions shall be established for conducting the tests in the specification and shall only be changed when they are the variables in the tests.

4.4 First article inspection. When first article inspection is required (see 3.2 and 6.2), samples shall be inspected as specified in table II. Failure of any sensor to meet the requirements of this specification shall be cause for rejection.

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TABLE II. First article inspection.

Inspection	Requirement	Method
Group I		
General examination	3.3, 3.4, 3.6 through 3.10	4.6
Accuracy	3.5.1	4.7.1
Repeatability	3.5.2	4.7.1
Sensitivity	3.5.3	4.7.3
Response	3.5.4	4.7.4
Warm-up time	3.5.5	4.7.5
Ripple	3.5.6	4.7.6
Supply voltage and frequency (steady-state)	3.5.7	4.7.7
Supply voltage and frequency (transient)	3.5.8	4.7.8
Temperature	3.5.13	4.7.13
Humidity	3.5.14	4.7.14
Enclosure	3.5.15	4.7.15
Salt Spray	3.5.16	4.7.16
Overpressure	3.5.17	4.7.17
Pressure cycling	3.5.18	4.7.18
Vibration	3.5.19	4.7.19
Shock	3.5.20	4.7.20
Burst pressure	3.5.21	4.7.21
Group II		
General examination	3.3, 3.4, 3.6 through 3.10	4.6
Accuracy	3.5.1	4.7.1
Repeatability	3.5.2	4.7.1
Insulation resistance	3.5.9	4.7.9
Electromagnetic interference emission and susceptibility	3.5.22	4.7.22

4.4.1 Sample size. Two sensors of each lot shall be subjected to first article inspection. One sample shall be subjected to the tests of Group I and one sample shall be subjected to the tests of Group II of table II. Each sample shall be supplied with the length of cable required for the intended application, but shall be not less than 100 feet.

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4.4.2 Inspection lot. An inspection lot shall consist of all the sensors of the same classification (see 1.2), produced under essentially the same conditions in the same facility from the same materials, and offered for delivery at the same time.

4.4.3 Order of inspection. The sample sensors shall be subjected to the inspections specified in table II in the order listed except that the steady-state supply voltage and frequency inspection (see 4.7.7) may be performed concurrently with temperature inspection (see 4.7.13). Any other deviation in the test order shall first be approved by NAVSEA.

4.4.4 Disposition of first article units. Samples subjected to first article inspection shall be considered consumed and nondeliverable as part of the contract.

4.5 Quality conformance inspection. Each lot of sensors offered for delivery shall be subjected to the inspections specified in 4.5.1 and 4.5.2.

4.5.1 Group A inspection. Each sensor in each lot offered for delivery shall be subjected to the group A inspection shown in table III. Failure of any sensor to meet the requirements of this specification shall be cause for rejection.

TABLE III. Quality conformance inspection.

Inspection	Requirement	Method
Group A		
General examination	3.3, 3.4, 3.6 through 3.10	4.6
Accuracy	3.5.1	4.7.1
Repeatability	3.5.2	4.7.1
Sensitivity	3.5.3	4.7.3
Group B		
Insulation resistance	3.5.9	4.7.9
Temperature	3.5.13	4.7.13
Enclosure	3.5.15	4.7.15
Overpressure	3.5.17	4.7.17

4.5.2 Group B inspection. Samples shall be selected in accordance with table IV. Each sample shall be subjected to the group B inspection specified in table III. Failure of any sensor to meet the requirements of this specification shall be cause for rejection.

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TABLE IV. Group B inspection sample size.

Lot or batch size	Sample size	Lot or batch size	Sample size
2 to 8	2	151 to 280	32
9 to 15	3	281 to 500	50
16 to 25	5	501 to 1200	80
26 to 50	8	1201 to 4200	125
51 to 90	13	4201+	3 per 100
91 to 150	20		

4.6 General examination. Each sensor shall be given a thorough examination to determine conformance to the requirements of this specification with respect to material, finish, workmanship, construction, assembly, weight, dimensions, and identification marking. Examination shall be limited to the examinations that may be performed without disassembly. Examination shall also include a check of all adjustments (see 3.3, 3.4, and 3.6 through 3.10).

4.7 Methods of inspection.

4.7.1 Accuracy and repeatability. The sensor shall first be flexed over its full pressure range by slowly increasing and decreasing the applied pressure for six continuous cycles. The reference measurement shall be made at not less than five equally spaced intervals over the entire range (both upscale and downscale). Precaution shall be taken to avoid overshoot. This procedure shall be applied three successive times to determine repeatability. Accuracy requirements of all reference measurements shall conform to 3.5.1. Repeatability shall conform to 3.5.2.

4.7.2 Reference measurement. A reference measurement shall be conducted when specified in the individual test. Measurements shall be made at not less than five equally spaced intervals over a one-time flexure over the entire sensor range both upscale and downscale. Precaution shall be taken to avoid overshoot. No adjustments to the sensor are permitted during the reference measurement.

4.7.3 Sensitivity factor. The sensitivity shall be determined as follows: Provide an input pressure to the sensor of 80 ± 5 percent of span. Measure the input pressure and corresponding electrical output. Increase the pressure by an amount not greater than 1 percent of span. Measure both the new pressure and corresponding new electrical output. Calculate the change in both applied pressure and electrical output as a percentage of span. Determine the ratio of electrical output percentage change to applied pressure percentage change. Repeat this procedure for a pressure decrease not greater than 1 percent of span. Performance shall conform to the requirements of 3.5.3.

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4.7.4 Response. A pressure ramp consisting of a pressure rise of not less than 40 percent of span occurring is not greater than 0.1 seconds shall be applied to the sensor. The maximum ramp pressure shall be maintained for not less than 0.50 seconds and shall not vary by more than plus or minus 2 percent of span. Performance shall conform to 3.5.4.

4.7.5 Warm-up time. The sensor shall be placed in an ambient temperature of 25 ± 2°C for not less than 2 hours to deenergize. Recording equipment and other auxiliary equipment shall be energized to ensure complete warm-up. An input pressure of 80 ± 5 percent of span shall be applied to the sensor and maintained constant during this test. The elapsed time between the application of line power and the point at which the sensor output reaches the conditions specified in 3.5.5 shall be measured. Performance shall conform to 3.5.5.

4.7.6 Ripple. Peak-to-peak ripple (the ac component of the dc output) shall be determined by connecting an oscilloscope to the output pins of the opto-electronics module. Ripple shall be determined at an input pressure of 80 ± 5 percent of span. Performance shall conform to 3.5.6.

4.7.7 Steady-state supply voltage and frequency. The sensor shall be operated for not less than 15 minutes in each of the conditions of normal, maximum, and minimum steady-state voltages and normal, maximum, and minimum steady-state frequencies (see 3.4.6). These conditions are labelled A through E and F through H for ac and dc powered sensors, respectively, in table V. The ambient temperature shall also vary for each of the conditions A through H. The sensor shall be operated for not less than 1 hour at each test temperature, 0 ± 2°C, 25 ± 2°C, and 65 ± 2°C, before the reference measurement (see 4.7.2). Test temperatures shall be accomplished by varying temperature in steps of 10°C each (30 minutes for each step) until the desired ambient temperature is reached. Performance shall conform to 3.5.7.

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TABLE V. Steady-state supply voltage and frequency.

	Voltage			Frequency		
Condition	Lower Limit	Normal	Upper Limit	Lower Limit	Normal	Upper Limit
(for ac)						
A		115			60	
B	107			58		
C			123	58		
D			123			62
E	107					62
(for dc)						
F	23.5			N/A	N/A	N/A
G		28		N/A	N/A	N/A
H			32.5	N/A	N/A	N/A

4.7.8 Transient supply voltage and frequency. Tests specified in 4.7.8.1 and 4.7.8.2 shall be conducted with a pressure input signal equal to 80 ± 5 percent of span. The sensor output shall be monitored throughout the test. Performance shall conform to 3.5.8.

4.7.8.1 Transient voltage. The electrical signal output shall be monitored at ambient temperature. Performance shall conform to 3.5.8.

4.7.8.1.1 Upper limit of steady-state voltage. With the sensor operating at the upper limit of steady-state ac voltage (see 3.4.6 and table V), a transient voltage of positive 15 volts recovering to the steady-state band in 2 seconds shall be superimposed. With the sensor operating at the upper limit of steady-state dc voltage (see 3.4.6 and table V), the dc powered sensor shall have a transient voltage of plus 2.0 volts recovering to the steady-state band in 2 seconds superimposed.

4.7.8.1.2 Lower limit of steady-state voltage. With the sensor operating at the lower limit of steady-state ac voltage (see 3.4.6 and table V), a transient voltage of negative 15 volts recovering to the steady-state band in 2 seconds shall be superimposed. With the sensor operating at the lower limit of steady-state dc voltage (see 3.4.6 and table V), the dc powered sensor shall have a transient voltage of minus 2.0 volts recovering to the steady-state band in 2 seconds superimposed.

4.7.8.2 Transient frequency (for ac powered sensors). The electrical signal output shall be monitored at ambient temperature. Performance shall conform to 3.5.8.

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4.7.8.2.1 Upper limit of steady-state frequency. With the sensor operating at the upper limit of steady-state frequency (see 3.4.6 and table V), a transient frequency of positive 1.5 Hz recovering to the steady-state band in 2 seconds shall be superimposed.

4.7.8.2.2 Lower limit of steady-state frequency. With the sensor operating at the lower limit of steady-state frequency (see 3.4.6 and table V), a transient frequency of negative 1.5 Hz recovering to the steady-state band in 2 seconds shall be superimposed.

4.7.9 Insulation resistance. The insulation resistance of the sensor shall be determined by applying 50 Vdc between electrical input and output circuits and between these circuits and ground. The temperature shall be 25 ± 5°C and the relative humidity shall be 50 ± 10 percent. The insulation resistance measurement shall be made immediately after a 2 minute period of uninterrupted test voltage application. If the indication of insulation resistance conforms to the specified limit (see 3.5.9) and is steady or increasing, the test may be terminated before the end of the 2 minute period. The sensor shall conform to 3.5.9.

4.7.10 Power interruption. An input pressure of 80 ± 5 percent of span shall be applied to the sensor and maintained constant during the test. With the sensor operating within the steady-state tolerances of voltage and frequency, the external power supply shall be interrupted for an interval of 3 to 4 seconds. The power supply shall then be reestablished to within steady-state tolerances. The sensor shall be operated at steady-state power for one minute. The power supply shall then be interrupted for an interval of 30 seconds. This cycle shall be repeated three times. The sensor shall conform to 3.5.10.

4.7.11 Short circuit. An input pressure of 80 ± 5 percent of span shall be applied to the sensor and maintained constant during the test. The sensor shall be deenergized. The electrical output pins of the opto-electronics module shall be connected together with no load resistance. The sensor shall be energized for five minutes. Immediately following the five minute period, the output pins shall be unshorted. The sensor shall conform to 3.5.11.

4.7.12 Line voltage reversal (dc). An input pressure of 80 ± 5 percent of span shall be applied to the sensor and maintained constant during the test. The positive 28 V dc signal shall be applied to connector pin "B". The dc reference signal shall be applied to connector pin "A". The sensor shall be energized for a period of ten minutes and then disconnected. The power shall then be correctly applied (pin "A" positive, pin "B" negative). A reference measurement shall be made (see 4.7.2). The sensor shall conform to 3.5.12.

4.7.13 Temperature. The sensor shall operate normally (without alignment or adjustment) throughout the following temperature cycle. Tolerances in operating characteristics shall be as specified herein. Performance shall conform to 3.5.13.

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(a) Hold test temperature at 0 ± 2°C for not less than 24 hours. During the last hour of operation, a reference measurement (see 4.7.2) shall be made.
(b) Increase test temperature in steps of 10 degrees each, at 30 minutes for each step, until plus 65 ± 2°C is reached and hold at that temperature for not less than 24 hours. During the last hour of operation, a reference measurement (see 4.7.2) shall be made.
(c) Reduce test temperature in steps of 10 degrees each, at 30 minutes for each step, until plus 25 ± 2°C is reached and hold at that temperature for not less than 24 hours. During the last hour of operation, a reference measurement (see 4.7.2) shall be made.

4.7.14 Humidity. Sensor shall be subjected to the following conditioning and test. The equipment shall be energized throughout the test. Performance shall conform to 3.5.14.

4.7.14.1 Conditioning. In order to establish a reference condition for the measurement of operating parameters and a valid basis for comparison of the effects of the conditioning to follow, the complete equipment shall be dried at a temperature not less than 40°C or more than 50°C for not less than 2 hours.

4.7.14.2 Reference measurements. Following the conditioning specified in 4.7.14.1, a reference measurement (see 4.7.2) shall be conducted at 25 ± 5°C and 50 ± 5 percent relative humidity. Sensor performance shall conform to 3.5.1.

4.7.14.3 Temperature cycling. The sensor shall then be subjected to four 24-hour cycles of temperature variation consisting of 18 hours at 65 ± 5°C and 6 hours at 25 ± 5°C. The relative humidity shall be maintained above 95 percent during the steady-state conditions. The transitions between temperatures shall be accomplished within the 6-hour period so that the time at the high temperature is not less than 18 hours. Each transition shall be not greater than 1 hour if the sensor remains in the chamber, or 15 minutes if a two chamber method is employed. The relative humidity need not be controlled during the transition periods.

4.7.14.4 Measurement during cycling. During the second cycle, the measurements required in 4.7.14.2 shall be performed at the end of the high temperature period (65 ± 5°C) with the sensor remaining in the chamber, before the temperature is decreased to 25°C. The sensor shall be energized for as brief a period as required to complete the measurements. A warm-up period may be permitted where previous tests indicate a definite period is required for the sensor to attain thermal stability.

4.7.14.5 Measurements after temperature cycling. After the four complete cycles, the measurements required in 4.7.14.2 shall be performed at 25 ± 5 °C with the sensor remaining in the chamber. In addition, the sensor shall be examined to detect evidence of physical degradation such as corrosion of metal parts or distortion of plastic, glass, or silicon parts.

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4.7.15 Enclosure. The sensor head and opto-electronics module (Type A only) shall be subjected to the splashproof test in accordance with MIL-STD-108. At the completion of the test, a visual examination shall be performed. Performance shall conform to 3.5.15.

4.7.16 Salt spray. A reference measurement (see 4.7.2) shall be made before the salt spray test. The sensor head and opto-electronics module (type A only) shall be subjected to the salt spray test in accordance with MIL-STD-202, method 101. Duration of the test shall be 96 hours. If the sensor is not designed for seawater service, the inlet pressure connections shall be plugged. A reference measurement shall be made (see 4.7.2). Performance shall conform to 3.5.16.

4.7.17 Overpressure. Before the overpressure test, a reference measurement (see 4.7.2) shall be made. The sensor shall be subjected to a pressure equal to 200 percent of the upper limit of its range for a period of 30 minutes. Within 10 minutes after release of this pressure, a reference measurement (see 4.7.2) shall be made for comparison. Performance shall conform to 3.5.17.

4.7.18 Pressure cycling. Before performing the pressure cycling test, a reference measurement (see 4.7.2) shall be made. The test shall be conducted by applying a periodic pressure change of not more than 20 percent of span to not less than 80 percent of span for a total of 260,000 cycles. The rate of cycling shall be within the range of 0.25 and 2.0 Hz. The sensor shall be energized throughout the test. After completion of the pressure cycling test, a reference measurement (see 4.7.2) shall be made for comparison. Performance shall conform to 3.5.18.

4.7.19 Vibration. Before the vibration test, a reference measurement (see 4.7.2) shall be made. The sensor shall be tested in accordance with MIL-STD-167-1 type I (environmental) vibration except that the upper frequency shall be 175 Hz; the amplitude (single) of vibration shall be in accordance with table VI; and for the variable frequency portion, the vibration level shall be maintained for 2 minutes at each integral value of frequency. If no resonances are observed, the 2 hour endurance test shall be conducted at 175 Hz. During the vibration test, a fluid pressure of 80 ± 5 percent of span shall be applied to the sensor. Sensor output during the test shall be monitored. Performance shall conform to 3.5.19.

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TABLE VI. Amplitudes of vibration

Frequency Range (Hz)	Table Amplitude (Single) for Exploratory (inches)	Table Amplitude (Single) for Variable Frequency (inches)
5 to 20	0.010 ± 0.002	0.030 ± 0.006
21 to 50	0.004 ± 0.001	0.020 ± 0.004
51 to 100	0.0015 ± 0.0003	0.010 ± 0.002
101 to 175	0.0005 ± 0.0001	0.0015 ± 0.0003

4.7.20 Shock. Before the shock test, a reference measurement (see 4.7.2) shall be made. The shock tests shall be conducted in accordance with grade A, class 1, type C using bulkhead mounting fixture 4-A in accordance with MIL-S-901. During the test, a fluid pressure of 80 ± 5 percent of span shall be applied to the sensor. Sensor output during the test shall be monitored. A post-test calibration shall be performed using adjustments provided. Performance shall conform to 3.5.20.

4.7.21 Burst pressure. The sensor shall be subjected to a liquid pressure equal to 300 percent of the upper limit of its range applied to the sensor for a period of 10 minutes. The sensor shall conform to the requirements of 3.5.21. No performance test shall be required after the burst pressure test. A reference measurement (see 4.7.2) shall be measured for information purposes.

4.7.22 Electromagnetic interference (EMI) emission and susceptibility. EMI tests shall be in accordance with the test methods specified in MIL-STD-462. Performance shall conform to 3.5.22.

4.8 Inspection of packaging. The sampling inspection of the preservation, packing, and container marking shall be in accordance with the requirements of section 5 and the documents specified herein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 Packaging requirements. Sensor equipment shall be individually preserved level A, C, or commercial, packed level A, B, C or commercial as specified (see 6.2) and marked including bar coding in accordance with MIL-E-17555 (see 6.2). Unless otherwise specified (see 6.2), method III shall apply for level A preservation.

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5.2 Marking. In addition to any special marking required (see 6.2), unit and intermediate packs and shipping containers and palletized unit loads shall be marked in accordance with ASTM-D-3951. Oxygen service sensors shall be marked "Oxygen Service" on their individual unit pack.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Sensors are intended to provide control input and monitoring of absolute pressure for shipboard engineering plants.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of sensors required.
(c) Sensor classification(s) required (see 1.2).
(d) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(e) When first article inspection is required (see 3.2 4.4, and 6.5).
(f) Opto-electronics module type A requirements (see 3.4.2.1) or type B requirements (see 3.4.2.2). If type B, specify requirements.
(g) Cable length required (see 3.4.3).
(h) Output (see 3.4.7) If optical, specify requirements (See 3.4.7.3). If digital, specify requirements and data format (see 3.4.7.4).
(i) Electrical connectors for digital output (see 3.4.8).
(i) When mounting is by pressure connection (see 3.4.15).
(j) If responsibility for inspection equipment shall be other than as specified (see 4.1).
(k) Level of preservation and level of packing required (see 5.1).
(l) Applicable acquisition requirements of MIL-E-17555 (see 5.1).
(m) When method III does not apply for level A preservation (see 5.1).
(n) If special marking is required (see 5.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
Appendix	DI-DRPR-80651	Engineering drawings	Level 3
4.4	DI-NDTI-80809	Tests/Inspection Reports	----

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The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Controls List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards which have been cleared and listed in DOD 5010.12-L (AMSDL) must be exhibited on a separate CDRL (DD Form 1423), included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors; that is, whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.2); and the number of items to be tested, as specified in 4.4 The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract. When ordering spare parts or repair parts for the equipment covered by this specification, the spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.7 Subcontracted material and parts. The packaging or preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Subject term (key word) listing.

Opto-electronics module
Sensor head

Preparing activity:
Navy - SH
(Project 6040-N002-01)

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ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix contains the format and content preparation instructions for the development of and revision to a conformance verification drawing. It is not intended that each requirement contained herein should be applied to every type of instrumentation. Portions of this appendix are subject to deletion tailoring depending upon the material, construction, and principle of operating requirements that are specified in the individual instrumentation specification or acquisition document. This appendix is applicable only when data item description DI-DRPR-80651 is cited on the DD Form 1423. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY
MIL-I-45208 - Inspection System Requirements.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

20.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
Y14.1 - Drawing Sheet Size and Format. (DOD adopted)
200 - Standard Reference Designations for Electrical and Electronics Parts and Equipments. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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30. CONFORMANCE VERIFICATION DRAWING FOR INSTRUMENTATION

30.1 Purpose. The conformance verification drawing contains the information necessary to verify that the instrumentation meets the requirements specified in the applicable instrumentation specification and acquisition document.

30.2 Content and format. One conformance verification drawing shall be developed for each specific type of instrumentation. It shall include all ranges, sizes, connections and other variations. The conformance verification drawing shall include the following minimum information (except as specified in 10.1) and shall be developed to the format in 30.3 through 30.9. A complete symbol list shall accompany all drawings and schematics.

30.3 Descriptive data.

(a) Instrument identification numbering system for instrumentation. This numbering system shall include, but may not be restricted to, the classification variables.
(b) Instrumentation identification number system for replaceable parts.
(c) Size, operating data, ranges, scale markings, and other data for proper selection.
(d) Test approval data, presenting the following information in tabular form:
    (1) Specification classification (unique alphanumeric variables to designate instruments).
    (2) Test report number and date.
    (3) Facility where test was conducted.
    (4) Authorized Government activity (NAVSEA) approval letter and date.
(e) A statement that instrumentation is in accordance with the requirements of the applicable instrumentation specification or acquisition document and to referenced specifications.
(f) Conformance verification drawing acceptance data, presenting the following information in tabular form:
    (1) Authorized Government activity (NAVSEA) acceptance letter and date.
    (2) Revision number.

30.4 Details of construction.

(a) Two or more representative assembly views, as required, to show clearly the details of the design, construction, and assembly of the instrumentation and to identify each part and its location. Identification of parts shall correspond to the list of materials. Assembly shall show how all mechanical parts are joined or attached.
(b) Sectional views or notes as necessary to show internal details.
(c) Details such as entrance provisions, gaskets, fastening techniques, welding symbols, mounting requirements, and other details as applicable.

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(d) Module enclosure requirements for separately mounted modules.
(e) Dimensions required to ensure interchangeability.
(f) Schematic of indicating system.
(g) Description of the nature and purpose of any adjustments.
(h) Welding procedures with acceptance data, including acceptance letter, report numbers and dates.
(i) Any special features.
(j) Complete weight.
(k) Location, size, and type of connections.
(l) Identification of quality control documents which show conformance with MIL-I-45208 or the quality control specification that is listed in the applicable instrumentation specification or acquisition document. The acceptance letter shall be referenced.

30.5 Parts list. The following information shall be presented in tabular form:

(a) Item number (corresponding to flag number identifying the part on the conformance verification drawing).
(b) Quantity of each part required per assembly.
(c) Name of part with sufficient information to readily identify the part (for example, screws: thread size, length and type of head, shall be specified).
(d) Material of part.
(e) Material specification (military, federal, or non-Government document number or Government activity (NAVSEA) drawing number).

NOTE: When substitution of a material specification is made, it is the responsibility of the contractor to provide written documentation to substantiate that the substituted material is equivalent to the specified material.

(f) Type, class, grade, size, military designation, or other classification of any referenced specification.
(g) Part number or identification assigned by assembly supplier.
(h) Name of actual manufacturer of part (when applicable).
(i) Part number or identification assigned by part supplier (when applicable).
(j) Onboard repair parts. Parts that are appropriate for, or are supplied as, onboard repair parts shall be indicated by a symbol in this column.
(k) Remarks column. Finishes, platings, or coatings along with the applicable specification or other requirements should be specified in this column.

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30.6 Table for special tools. Special tools required for the instrumentation shall be presented in the following tabular form:

(a) Item number (corresponding to flag number identifying the tool on the conformance verification drawing).
(b) Quantity of each tool required per instrument.
(c) Name (description) of tool (include generic name).
(d) Tool specification (military, federal, or non-Government document number or Government activity (NAVSEA) drawing number).
(e) Tool number or identification assigned by assembly supplier.
(f) Name of actual manufacturer of tool (when applicable).
(g) Tool number or identification assigned by tool supplier (when applicable).
(h) Description of tool's application.
(i) Remarks column. Special techniques or other usage requirements shall be explained in this column.

30.7 Electrical system schematics. Electrical schematics shall be included in the verification conformance drawing. A complete schematic shall contain all the parts in the fluid or electrical systems. When confusion over its function could result, a simplified schematic containing only the major components shall also be provided. This simplified schematic may be presented in block diagram format.

(a) Complete electrical system schematic. A single schematic shall represent clearly the operation and the function of the electrical circuitry within the instrumentation. The schematic shall contain all parts (including components, connectors, alarms, and so forth) which make up the electrical system. The following features shall be incorporated into the schematic.
    (1) In preparation of the schematic, emphasis shall be placed on simplicity and ease of understanding of circuit operation. Physical placement of components and connecting wiring may be ignored in the interest of simplicity and clarity of this diagram.
    (2) A thin, broken line shall be used to represent the boundaries of each unit or subassembly. Terminals, to which external connections are made, shall be shown within these boundaries, with the numbers, markings, type of signal, power and ground, as appropriate.
    (3) Each part (such as resistors, capacitors, relays, and so forth) shall be given a unique reference designation consisting of a letter denoting the type of part (as required by ANSI 200) and a number assigned consecutively. The numbers shall be assigned in a logical sequence of electrical current or signal flow through the circuit.
    (4) In addition to where the reference designation, parts not conforming to a military specification, where permitted, shall have the following information noted adjacent to the part.

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Where numerical values are given, a code shall be noted to designate the units used:

a. Resistors - resistance, power rating, and tolerances. If variable, an arrow to indicate clockwise rotation of the control shaft.
b. Capacitors - capacitance, voltage rating, and tolerance.
c. Reactors - inductance and voltage rating.
d. Semi-conductors - type number (JEDDC number is adequate).
e. Integrated circuits - operational symbol diagram of input-output relationship, terminal numbering corresponding to a representative schematic, and type number.

(5) Supply voltages, phases, and frequencies and transformer terminal voltage shall be indicated and labeled as to purpose.
(6) Table for troubleshooting of electrical system. The content table shall include each test point, as identified on the electrical system schematic, with voltage, waveform or other electrical parameter that should be measured at each test point.

30.8 Selection and installation considerations.

(a) Performance data.
(1) Accuracy (in percent of span).
(2) Shock and vibration classification.
(3) Degree of water tightness of the enclosure.
(4) Electromagnetic interference and pulse susceptibility.
(b) Dimensional outline of the instrumentation showing overall and principle dimensions in sufficient detail to establish space requirements in all directions necessary for installation, servicing, exclusive of space required for operator observation of the indication.
(c) Special considerations which may affect selection or installation.
(1) Ambient temperature range.
(2) Calibration points and adjustments.
(3) Orientation.
(4) Location of instrumentation relative to vibrating equipment.
(5) Protection of the instrumentation from pulsations and spikes in the parameter being measured.
(6) Selection of the instrumentation range relative to the operating range of the system.
(7) Application for each type connection.
(8) Cleaning procedure or reference to the cleaning procedure used.
(9) Selection of the instrumentation for compatibility (materials, temperature, pressure, and so forth) with the ambient environment and with the parameter being measured.

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30.9 Drawing format.

(a) Unless otherwise approved by the authorized Government activity, a maximum of three sheets shall be allotted for single functioned system instrumentation and a maximum of ten sheets for instrumentation containing a multiple functioned system. A single functioned system is one that performs only one operation such as: expands or contracts a pressure elastic element, steps up or steps down the voltage, conditions one electrical signal, or winds or unwinds a bimetallic element. A multiple functioned system is one that contains two or more single functioned systems.
(b) Each sheet shall be zoned.
(c) Title block shall be included on each drawing sheet and shall include the following information:
  (1) Title, drawing number, and revision letter. Each sheet shall contain the same title, drawing number, and revision letter.
    a. Title - the title shall consist of the name by which the instrumentation is known.
    b. Drawing number - the drawing number shall consist of alphanumeric characters which may be separated by dashes or slashes. The total number of characters in the drawing number (including dashes and slashes) shall be not greater than 15. Blank spaces are not permitted within the drawing number.
    c. Revision letter - the revision letter shall denote the latest approved version of the drawing. The revision letter of conformance verification drawing shall not be changed until all the changes under that revision have been accepted in writing by the authorized Government activity (NAVSEA). No changes made to the conformance verification drawing shall be considered a revision until after the initial version of the conformance verification drawing has been accepted in writing by the authorized Government activity. After initial submittal of the conformance verification drawing, no changes shall be made during the initial drawing review process unless the change is either requested by the authorized Government activity or the change is documented in written correspondence by the contractor.
  (2) Sheet _____ of _____.
  (3) Tolerance on dimensions for fractions, decimals, and angles. Units of the dimensions specified on the conformance verification drawing.
  (4) Contractor acceptance block (appropriate signatures and dates).
  (5) Commercial and Government entity (CAGE) code for manufacturer.
  (6) Scale.
  (7) Reference drawings.
  (8) Manufacturer's name and address.
  (9) Drawing size.

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(d) A revision block shall be included on each sheet of the conformance verification drawing and shall contain the following information in tabular form:
(1) Revision letter.
(2) Description of revision.
(3) Acceptance letter serial number and originator identification.
(4) Acceptance data.
(e) Sheet size and format not specified herein shall be in accordance with ANSI Y14.1.
(f) Classification designations. No Government security classification designation such as confidential or secret shall appear on the conformance verification drawing unless a particular classification is specified by the Government.

40. CONFORMANCE VERIFICATION DRAWING ACCEPTANCE

40.1 Acceptance. Acceptance shall be granted by the authorized Government activity only after the conformance verification drawing is found to meet all the requirements specified in 20. through 40.1.

50. INSTRUCTIONS

50.1 Acquisition document instructions. The acquisition document should contain provisions that address submission, review, extension, disapproval, default, acceptance, and waiver of conformance verification drawings in addition to the effects on the delivery schedule due to delays in conformance verification drawing acceptance. NAVSEA shall be designated as the activity that accepts or disapproves the conformance verification drawing.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-24796(SH)	95/06/22

3. DOCUMENT TITLE

SENSORS, ABSOLUTE PRESSURE, FIBER OPTIC (NAVAL SHIPBOARD USE)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY TECHNICAL POINT OF CONTACT (TOPC):

a. NAME	b. TELEPHONE (Include Area Code)	
Mr. Frank LeLand SEA 03J4 ALL CORRESPONDENCE SHOULD BE ADDRESSED TO:	(1) Commercial (703) 602-6137	(2) AUTOVON 332-6137
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command SEA 03R42, 2531 Jefferson Davis Hwy Arlington, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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